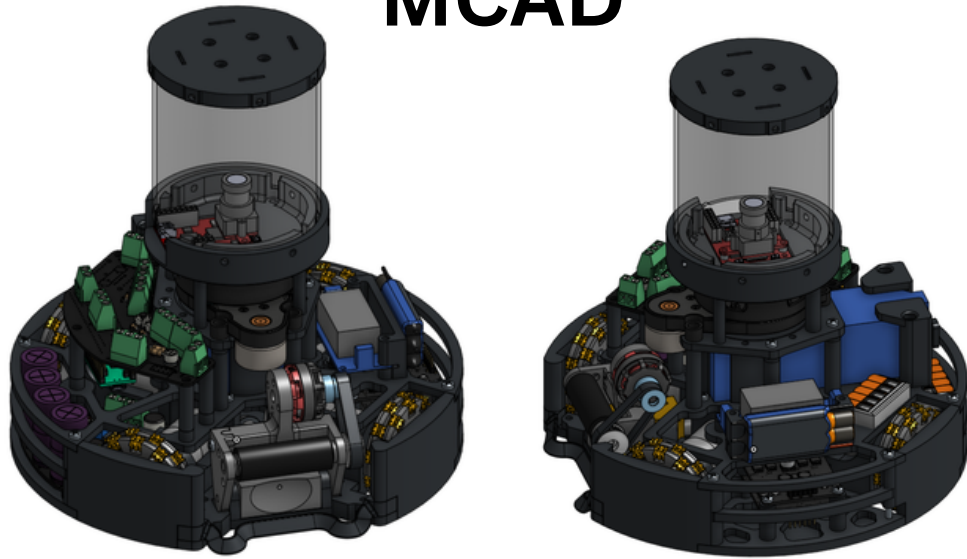


MCAD

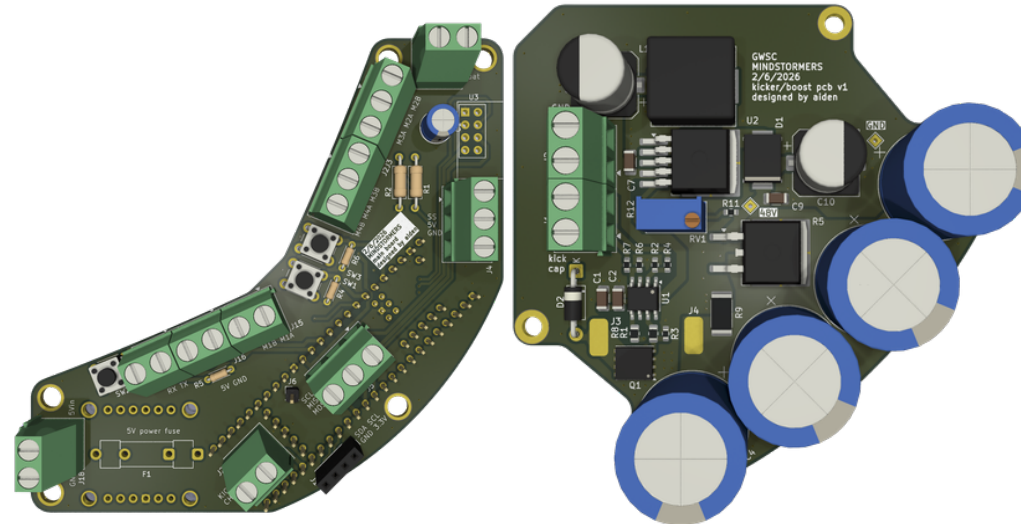


Our 2nd year in the RCJA Open Soccer competition, we spent an extended period of time developing the robot's CAD in Onshape.

Improvements to previous editions:

- Implementing weight reduction using FRC pocketing techniques
- Greater modelling of components prior to design
- Utilising 'Krayon CAD' to verify layouts and design choices
- Use of Onshape versioning to track development and progress
- Modular design with easily accessible electronics and wheels

ECAD



The greatest development in the ECAD this year was the use of custom PCBs from JLCPCB. 3 different designs were made, however only the mainboard (left) and Kicker were purchased. As discussed in the Developments and Innovations, the Kicker PCB includes a 48V boost and extremely fast discharge through mosfets chosen for ultra-low gate charge and ultra-fast rise times. The mainboard multiplexes the Teensy 4.1 microcontroller to all the different components, such as the BNO08x IMU, Cytron MDD3A motor drivers, OpenMV RT1062 camera and nRF24L01 wireless communication board.

Software

Using the Teensy 4.1 as the robot's processor, we could either use the Arduino IDE or another software to program it. In the end, we chose to use VS Code with PlatformIO, which allowed us to create more complex algorithms faster and to compile them much quicker, saving time and energy.

Code structure:

- Using PlatformIO, we can separate different subsystems into different files, allowing for less complexity and a more efficient work flow
- 'Include' files enable different files to use information from other subsystems, ensuring easy access between APIs
- Each main loop, the robot:
 - Processes data from the OpenMV and sensors
 - Updates the localisation
 - Determines the strategy to follow
 - Communicates all this to the other robot
 - Tells motors to start moving

Developments + learning:

This year, we moved to the Teensy and the PlatformIO system. Compared to last year, we learnt how different hardware buffers, DMAs and hardware serial processes interact. One of the newest developments was the use of AI to automate annoying tasks!

Developments + innovations

Our primary two innovations in the making of our robots are the high power Kicker circuit board and custom spiky omniwheels.

The kicker was designed in KiCAD, utilising a XL6019 boost converter for 4S (~16V) to 48V boost, and an CSD19534Q5A mosfet with a UCC27624 gate driver. The solenoids were also rewound, achieving a maximum kick speed of over 6 m/s.

After over 20 iterations of different dimensions, designs and 3D print settings, the final spiky omniwheels printed on a Ender-3 S1-pro in PLA were very cheap, only requiring small metal dowel pins, ensuring smooth, high grip movement,



Jared Jin: states software

Aiden Blanchfield: hardware, electrical, software

Issues + challenges

The greatest challenges faced in the robots were finding the issues in the Kicker PCB circuit and mitigating deficiencies of the OpenMV RT1062.

The Kicker:

- 5 PCBs were ordered after several months of design and checking
- All PCBs boosted the voltage exactly as specified
- However, the 2nd PCB worked for a little while, but then died
 - The initial reasoning was that the wires had shorted and killed the board
- Then, it happened again, after significant short circuit protection
- In the end, the issue was determined to be the flyback diode for the solenoid (1N4007) being **too slow** to dissipate the voltage across the solenoid, shorting the main mosfet.
- The solution was to reduce the voltage and order faster diodes!

The OpenMV RT1062:

- The RT1062 seemed like a worthy vision processor for the robots due to initial lower costs and lower latency compared to other systems
- However, after we spent the budget on them, we found that they could not track the goal and the ball at the same time
- Thus, the goals were removed from the vision code, and only the ball was tracked. The robot was instead localised using the LiDAR

Strategy

This year, with the purchase and use of a new 360° LiDAR, we chose to prioritise localising the robot and completing manoeuvres using poses and coordinate goals. This allows us to undertake more complex strategies such as hiding the ball, avoiding opponents and swapping positions.

Additionally, with the use of the nRF24L01 wireless communication module, we designed our own custom wireless transmission protocol to update robot state and locations, enabling the other robot to find the ball even if it is not in sight.

Finally, we also use the LiDAR alongside point-cloud transformations to identify areas on the field that do not conform to the walls. These areas are then labelled as opponent robots, and after filtering, allow us to track the opponent.

